

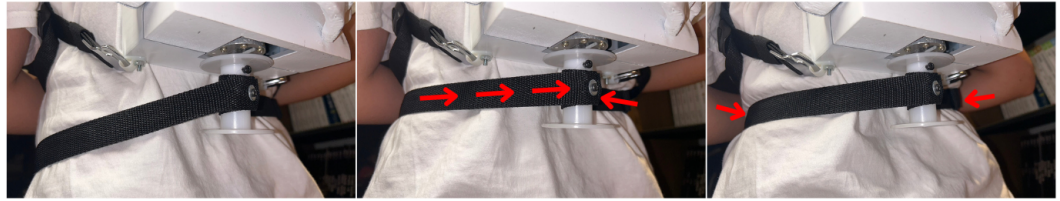
EmBelt: A Haptic Device for Extended Reality Experience to Better Understand Body Image Concerns-Related Eating Disorders

Hsuan Hui Yi
National Tsing Hua University
Hsinchu, Taiwan
vankleyyi@gapp.nthu.edu.tw

Yi Chun Ko*
National Tsing Hua University
Hsinchu, Taiwan
ko.yichun@gapp.nthu.edu.tw



(a) System overview.



(b) Belt section's expansion and contraction.

Figure 1: (a) EmBelt is a motor-driven haptic system that applies adjustable compression to the waist. (b) When the belt is loose, it symbolizes a relaxed part of the story. As the narrative becomes oppressive, the belt tightens, driven by a motor rotating the roller, exerting pressure on the audience's waist.

ABSTRACT

This study introduces EmBelt, an innovative haptic device integrated with VR storytelling that creates an extended reality (XR) experience designed to enhance understanding of eating disorders (EDs) related to body image concern (BIC) among individuals without such anxieties. The system dynamically adjusts compression levels in response to user interactions, providing a portrayal of the pressures linked to the thin ideal and extending related topics. Future directions include potential platform expansion to reach a broader audience and applications in medical treatment, underscoring EmBelt's potential as a tool for various intervention uses.

KEYWORDS

Haptics, Eating disorders, Body image concerns, Extended reality

ACM Reference Format:

Hsuan Hui Yi and Yi Chun Ko. 2024. EmBelt: A Haptic Device for Extended Reality Experience to Better Understand Body Image Concerns-Related Eating Disorders. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Emerging Technologies (SIGGRAPH Emerging Technologies '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641517.3665785>

*Corresponding author

1 INTRODUCTION

Eating disorders (EDs) are behavioral conditions marked by disrupted eating behaviors and distressing thoughts, impacting physical, psychological, and social well-being, with potential for severe outcomes including suicide [American Psychiatric Association 2022]. The causes and symptoms of EDs are complex, but body image concern (BIC) is a significant diagnostic feature [Baker et al. 2023; Mallaram et al. 2023].

BIC is a multidimensional internal evaluation of one's body [Littleton et al. 2005], leading to harmful body relationships. Amongst different components of manifestations, behavioral disturbances are at the core of attitudinal disturbances of body image, which include body checking and body avoidance. Body checking includes actions like frequent weighing, mirror checking, and scrutinizing perceived flaws [Linardon et al. 2019].

The "thin ideal" is a prevalent issue influenced by cultural preferences for slender physiques [Melisse et al. 2020], leading to complex and potentially harmful body image relationships [Voelker et al. 2015]. Wearing tight-fitting attire can cause physical discomfort and illness [Parazzini et al. 1995; Wai et al. 2016].

To address these challenges, we developed EmBelt, a haptic device integrated with VR storytelling. This device simulates the physical and psychological pressures of the thin ideal, while VR depicts body-checking behaviors characteristic of BIC, aiming to enhance understanding of EDs and BICs among those unfamiliar with these conditions.

2 PROTOTYPE

The system is composed of three primary components (Figure 2):

Electronic Components. This part consists of a direct current (DC) motor, Arduino UNO Board, and a 12 V power supply. The DC motor

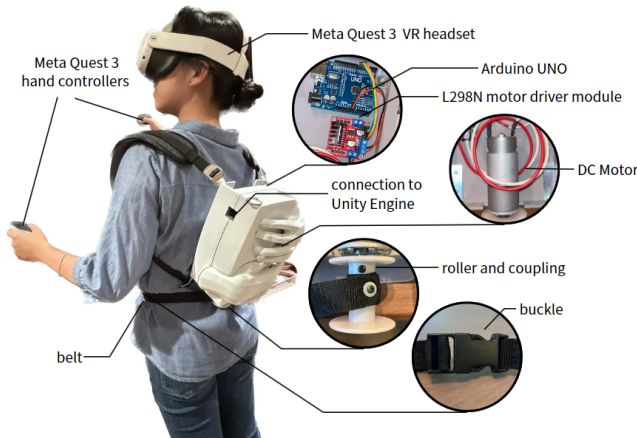


Figure 2: The wearing and internal structure of EmBelt. The prototype version is used for illustration purposes.

boasts a constant torque load of $3.12 \text{ N} \cdot \text{m}$ and can adjust rotational speed, direction, and duration, affecting the pressure on the player.

Mechanical Components. The compression belt, made of 2-inch wide low-elasticity nylon, applies pressure to the waist and accommodates sizes up to 52 inches. A roller and coupling mechanism ensure uniform pressure, and a quick-release buckle provides safety.

Structural Components. A box protects the electronic components, while shoulder straps enhance portability and positioning. The box exterior is designed to convey a sense of constraint and attract curiosity, promoting the experience.

Synchronization. The Unity Engine synchronizes with potential media options like VR HMD, Tablet, and Projection Mapping Systems, and sends signals to the Arduino, which controls the DC motor. The waist compression device adjusts speed and intensity based on the body shapes of characters in the animation. (Figure 3)

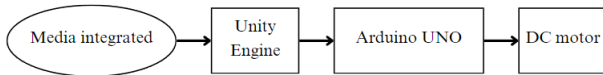


Figure 3: System architecture diagram. Explaining how platforms are connected.

3 DEMONSTRATION

The prototype evolves into a demonstration version by refining EmBelt casing design, enhancing motor precision, and emphasizing VR interaction for immersive experiences. An animation visually depicts the mental state of ED and BIC behavioral manifestations, symbolized by alterations in body shape. For an optimal immersive experience, viewers are recommended to use a VR headset. The animation portrays a black backdrop, representing the universality of

these experiences, with viewers engaging in a first-person perspective, observing their reflection in a mirror. Information about ED prevalence and statistics concludes the experience. The 10-minute session includes equipment fitting, XR experience with synchronized haptic feedback, and a post-removal discussion encouraging reflection on the experience.

4 FUTURE WORK

This study could develop in the following areas:

Multi-User Device Interaction and Platform Expansion. To spread awareness about ED to a wider audience, the device can connect multiple VR headsets for enhanced interaction. Expanding VR presentations to mobile phones and computers will increase accessibility.

Treatment of ED or Related Disorders. VR and XR have proven effective in medical treatments [Chard and van Zalk 2022; Chen et al. 2022]. Collaborating with healthcare organizations can help apply this technology to treat ED and related disorders.

ACKNOWLEDGMENTS

We thank I-Hsuan Wu, Jen-Jen Chou, and Tzu-Hsuan Wu for their contributions to the VR animation prototype, and Ting-Han Wu for coding the demonstration version. We also appreciate the financial support from the National Science and Technology Council, the Ministry of Education of Taiwan, and NTHU's College of Arts and Office of Research and Development, College of Engineering.

REFERENCES

- American Psychiatric Association 2022. *Diagnostic and statistical manual of mental disorders (5th ed.)*. American Psychiatric Association.
- Sarah Baker, Christophe Maïano, Simon A. Houle, Lindsey Nadon, Annie Aimé, and Alexandre J.S. Morin. 2023. Profiles of body image concerns and their associations with disordered eating behaviors. *Appetite* 191 (2023), 1–11.
- Ian Chard and Nejra van Zalk. 2022. Virtual Reality Exposure Therapy for Treating Social Anxiety: A Scoping Review of Treatment Designs and Adaptation to Stuttering. *Frontiers in Digital Health* 4 (2022), 842460.
- Yuhan Chen, Zhuoren Zhou, Min Cao, Min Liu, Zhihao Lin, Weixin Yang, Xiao Yang, Denzel Dhaidhai, and Peng Xiong. 2022. Extended Reality (XR) and telehealth interventions for children or adolescents with autism spectrum disorder: Systematic review of qualitative and quantitative studies. *Neuroscience & Biobehavioral Reviews* 138 (2022), 104683.
- Jake Linardon, Matthew Fuller-Tyszkiewicz, Xochitl de la Piedad Garcia, Mariel Messer, and Leah Brennan. 2019. Do body checking and avoidance behaviours explain variance in disordered eating beyond attitudinal measures of body image? *Eating behaviors* vol 32 (2019), 7–11.
- Heather L Littleton, Danny Axsom, and Cynthia L S Pury. 2005. Development of the body image concern inventory. *Behaviour Research and Therapy* 43, 2 (2005), 229–241.
- Ganesh Kumar Mallaram, , Pragya Sharma, Dheeraj Kattula, Swarndeep Singh, and Poojitha Pavuluru. 2023. Body image perception, eating disorder behavior, self-esteem and quality of life: a cross-sectional study among female medical students. *Journal of Eating Disorders* 11 (2023), 2050–2974.
- Bernou Melisse, Edwin de Beurs, and Eric F. van Furth. 2020. Eating disorders in the Arab world: a literature review. *Journal of eating disorders* 8(1) (2020), 59.
- F Parazzini, M Marchini, L Luchini, L Tozzi, R Mezzopane, and L Fedele. 1995. Tight underpants and trousers and risk of dyspermia. *International Journal of Andrology* 18, 3 (1995), 137–140.
- Dana K Voelker, Justine J Reel, and Christy Greenleaf. 2015. Weight status and body image perceptions in adolescents: current perspectives. *Adolescent health, medicine and therapeutics* 6 (2015), 149–58.
- Karmen Wai, Philip Douglas Thompson, and Thomas Edmund Kimber. 2016. Fashion victim: rhabdomyolysis and bilateral peroneal and tibial neuropathies as a result of squatting in 'skinny jeans'. *Journal of Neurology, Neurosurgery & Psychiatry* 87, 7 (2016), 782–782.